KVKK-Bench: Measuring Personal-Data Masking in Turkish Legal Text, with Small On-Device Models

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Katip · https://github.com/fikrifirat/katip

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Abstract

General-purpose AI assistants are now part of legal work. The Turkish data-protection authority's 2026 guide for lawyers states that uploading client documents to such assistants can amount to a transfer of personal data abroad under art. 9 of Law No. 6698 (KVKK), and recommends masking personal data first. Masking needs a detector that works on Turkish legal prose and can run on the lawyer's own machine, but published Turkish PII models are evaluated on their own, mostly synthetic, data. We release **KVKK-Bench**, a public benchmark with a fixed scorer that ranks systems on real text only (Yargıtay sentences with held-out Turkish and foreign names, and Wikipedia sentences), reports a character-level *leak rate*, and closes a loophole by which over-masking scored perfect precision. We evaluate 14 systems and release two small models, **Katip Mask Base** (110M parameters, 111 MB int8) and **Katip Mask Nano** (2.5M, 2.9 MB, 50 pages/s on a laptop CPU), both of which run in a browser. Base scores 97.0 Overall [95% CI 96.5–97.5], 1.1 points above the best general model (XLM-R Large, 95.9) but with twice its leak rate (6.9% vs 3.4%). We do not claim a state of the art: the margins are small, part of the test is in-domain for our models, and no real client documents are labelled yet. The contribution is a shared, reproducible yardstick and a practical model family for local masking.

1 Motivation

Legal setting. Law No. 6698 (KVKK) governs personal data in Türkiye [29]. Since 1 June 2024, transfers abroad follow a GDPR-like tiered regime (adequacy decision, then appropriate safeguards such as standard contracts notified to the Authority, then narrow incidental exceptions; art. 9 as amended by Law No. 7499) with implementing rules in a July 2024 regulation [20]. In September 2026 the Authority published an application guide for lawyers, prepared with the Union of Turkish Bar Associations [21]. It treats lawyers as data controllers, notes that their exemption from the controllers' registry (Board decision 2018/32 [14]) does not exempt them from other duties, and has a section on generative AI. It names ChatGPT, Claude and Gemini and states that uploading client files or case documents containing personal data may constitute a transfer abroad under art. 9 when the provider, its servers or its sub-processors are abroad, and a domestic transfer under art. 8 otherwise (pp. 120–121). It recommends masking or anonymising personal data "as far as possible", not uploading special-category data, and sharing only the minimum needed (p. 146).

Enforcement. The guide quotes Board decisions that fined lawyers for unlawful processing, e.g. 2020/429 (phone numbers of a debtor's relatives and colleagues) [15], 2021/111 (50,000 TL, an SMS to a number known to belong to a relative) [16] and 2021/1111 (a witness's criminal record obtained without authority and used in a petition) [17]. It also quotes decisions finding no violation where a lawyer submitted personal or special-category data to a court for the client's case (2021/925, 2023/1414) [18, 19]. So the exposure is not court filing as such; it is sending client data to third parties without a legal basis, which is what pasting a file into a foreign AI service can be.

Why on-device. A remote masking service, or a connector run by the AI provider, moves the original document off the lawyer's machine before anything is masked. Only masking on the device keeps the originals local. That requires a detector that is (i) accurate on real Turkish legal prose, with court casing (*Ahmet YILMAZ*), case suffixes (*Kaya'nın*) and foreign clients' transliterated names, and (ii) small enough for a browser tab or a laptop. Masked text is still pseudonymised personal data, so masking reduces but does not remove KVKK obligations.

Gap. Several Turkish PII models are public, but each reports on its own data, usually synthetic or a validation split. Our own first model illustrates the risk: BERTurk fine-tuned on synthetic legal documents found every name in synthetic tests and 41% of names in real Wikipedia sentences [8].

Related work. Anonymisation benchmarks with real text are English: TAB on ECHR judgments [27] and i2b2 clinical notes [32]. Recent multilingual PII bench-

marks either generate records under controlled conditions (REDACT, 25 languages [35]) or pool existing corpora under one label set (PIIBench [13]), and an audit of widely downloaded masking models on semi-synthetic sentences found large gaps [31]. Thunder-DeID is the closest analogue in a single jurisdiction: a de-identification dataset and model for Korean court judgments [11]. Name detectors are known to fail unevenly across demographic groups [22], which motivates our foreign-name track. An independent multilingual evaluation also finds Presidio and an OpenAI Privacy Filter fine-tune well behind a trained encoder [10], consistent with our results. For Turkish, legal NER data exists [4] but its person names are mostly anonymised as "..."; Turkish legal encoders (Mecellem [33], HukukBERT [26]) are candidate base models we have not yet tried; and an audit found that 70% of commonly used Turkish benchmarks fail basic quality criteria [3], which is why we document construction, licences and known defects in detail (Appendix B).

2 KVKK-Bench

Real-text tracks (ranked). *court-turkish*: the 109 sentences of the TurkishLegalNER *test* split [4] that contain anonymised person slots, each slot filled three times (327 documents, 20,346 words, 1,011 names). Names come from Wikidata (CC0): given and family names of Turkish citizens, kept only if no word of the name appears in Katip's lexicons, any synthetic training set or Turkish Wiki NER train, so no system can score by memorising them. Casing follows court style (Title 43%, *Ahmet YILMAZ* 37%, all caps 20%) and the case suffix after each slot is re-inflected to agree with the new name. *court-foreign*: the same slots filled with names from held-out families of nine origin groups (Arab, Persian, Afghan, Georgian, Armenian, Greek, Slavic, Azerbaijani, Western) with real-world transliteration variants (327 documents, 1,011 names); the two court tracks are paired by slot. *prose*: the 1,000 sentences of the Turkish Wiki NER test split, PERSON spans only, as annotated by its authors [1] (654 names).

Generated tracks (check only). Petitions, contracts, correspondence, health records and court filings from Katip's generator (80 documents, 16 styles, seed 4242) and 300 generated petitions about foreign clients with foreigner IDs (YKN), passports and home-country IDs. They carry every non-name type (TCKN, VKN, IBAN, phone, e-mail, address, plate, birth date, health). Because Katip's models are trained on the same generator family (with different names), these tracks are shown as a regression check and are *excluded* from the ranking. A name-free legal-prose track (3 documents) counts false alarms.

Scorer. A gold value is *found* if a predicted span of the same type covers it, or lies inside it and covers at least half of it (and $\geq$3 characters; whitespace- and case-insensitive). A predicted span is *correct* if it overlaps a gold value of its type *and* at least half of its letters and digits lie inside such values. The second condition is a fix in v0.1.2: before it, one span over the whole document scored 100% precision, the same class of defect that benchmark audits report elsewhere (e.g. empty responses counted as successes [40]). Per dimension we compute F2 [34] from pooled recall and precision (a miss costs four times a false alarm). **Overall** is the weighted mean of three real-text name dimensions: Turkish names in court text (weight 25), foreign names in court text (10), names in general prose (10). The **leak rate** is the share of gold personal-data characters left unmasked by any predicted span, averaged over the three real tracks; a half-masked name counts as half a leak. We also report the clean-document rate and false alarms per 1,000 words.

Uncertainty. 95% CIs come from a percentile bootstrap [6] with 10,000 resamples that resamples the 109 court sentences as clusters (keeping their three fills and both court tracks together) and the 1,000 prose sentences independently. Comparisons with Katip Mask Base are paired (same resample for both systems); we report two-sided bootstrap p-values and correct across the 13 comparisons with Holm's method [12].

Release. Public split, scorer, runner, saved predictions of every system and the leaderboard Space are open [7, 8]; a canary string marks every file. Licences follow the sources: TurkishLegalNER annotations Apache-2.0 (court decisions carry no copyright under FSEK art. 31), Wiki NER CC BY-SA 4.0, generated documents CC BY 4.0. Every inserted value is synthetic.

3 Systems

Katip Mask Base · tr v2: BERTurk [30] fine-tuned for 3 epochs on 8,000 synthetic legal documents plus 17,967 Turkish Wiki NER *train* sentences (person names), tagging names, addresses and health terms; int8 ONNX. **Katip Mask Nano · tr v1.1**: a 2-layer BERT (hidden 128, 16k WordPiece) trained from scratch for 5 epochs (458 s on 4 CPU cores) on 8,000 synthetic documents, 4,182 real-text documents (Wiki NER train, TurkishLegalNER *train*-split sentences with filled names) and a transliterated foreign-name mix whose test families were held out. Both run inside Katip's rule engine (checksummed IDs, phones, IBANs, addresses, casing and suffix handling; 0.3 MB) with the browser's own code, detection threshold 0.5 and *trust* 0.5 (model spans scored at or above it keep their edges; chosen on dev data). **Baselines** (run unmodified, label maps written from each model's `id2label` before scoring): Presidio with spaCy `xx_ent_wiki_sm` [23]; Turkish NER (BERTurk NER, XLM-R Base Turkish NER); XLM-R Large fine-tuned on English CoNLL-03 [5]; GLiNER multi-PII [37] and GLiNER2-PII [38, 39] prompted with *person*; Turkish PII models (ModernBERT-TR PII, OpenMed PII

Turkish 33M); multilingual PII models (OpenAI Privacy Filter [25], OpenMed Privacy Filter v2, Piiranha); and Qwen2.5-1.5B-Instruct [28] prompted to list personal data as JSON. All systems receive identical text and are timed on one Apple-silicon laptop (12 CPUs) in Python; a page is 500 words.

4 Results

Main result (Table 1). Katip Mask Base has the highest Overall, 97.0. Its lead over XLM-R Large is 1.1 points [0.3, 2.1], over XLM-R Base Turkish NER 1.3 [0.5, 2.2] and over GLiNER2 2.1 [1.2, 3.0]; all 13 paired differences remain significant after Holm correction (largest adjusted $p = 0.008$). These margins are small, and the ranking depends on the metric: XLM-R Large leaks *half as much* (3.4% vs 6.9%, difference 3.5 points [2.5, 4.6]); the leak difference with XLM-R Base, GLiNER2, BERTurk NER and GLiNER is not significant (all $p > 0.1$). Of the systems we have run in a browser (only the two Katip models so far), Base is the stronger, at one twentieth of XLM-R Large's download size.

Small models. Nano reaches 93.3/93.1 F2 on Turkish and foreign names in court text at 2.9 MB and 50 pages/s, 29 times faster than Base, but collapses on general prose (57.8 F2, 44.5% leak): a sub-5 MB model covers legal register, not open-domain Turkish. Off-the-shelf multilingual PII models of 1.4B parameters score 51–57, below the 11 MB Presidio on prose and far below Turkish NER models on court text; a prompted 1.5B LLM is both slow (80 s per page on CPU) and unreliable (57.5, CI width 22 points).

Error analysis (Base). On court-turkish it misses 1 of 1,011 names; its 5 false spans are rapporteur names whose span also swallowed the title (*Tetkik Hakimi* ...), so the name is masked but the span fails the 50% rule. On court-foreign it misses 30 of 1,011; recall by origin is 95–100% except Azerbaijani names (82%, 41/50), which are often built from Turkish word stems (XLM-R Large: 98%); per-group gaps of this kind are the demographic failure mode reported for English PII masking [22]. On prose it misses 67 of 654: single-token names 11% (32/300) and names of three or more tokens 27% (22/81), versus 5% for two-token names; false spans are mostly titles of works and organisations (*Bohemian Rhapsody*, *Florya Metin Oktay*). The higher leak rate of Base versus XLM-R Large comes from prose (17.6% vs 9.8%), i.e. long and single-token names in open-domain text. XLM-R Large's false spans are mostly lone punctuation tagged PER (harmless for privacy).

Generated tracks (not ranked). With Katip's rules, Base and Nano find every TCKN, phone, e-mail and address in the generated documents, and Base finds 95.7 F2 of health terms. Name-only systems score 0 on these types by construction. One harness defect: the v0.1.2 run of the Katip rows used a rule file without the YKN, passport and foreign-ID recognisers shipped in the app, so both rows score 0 on those 557 values ("Other IDs" 16.5). The shipped rules alone find 100% of YKN and passport numbers and 96% of foreign IDs on that track. Because these are generated documents, Overall is unaffected; the rows will be re-scored in v0.1.3.

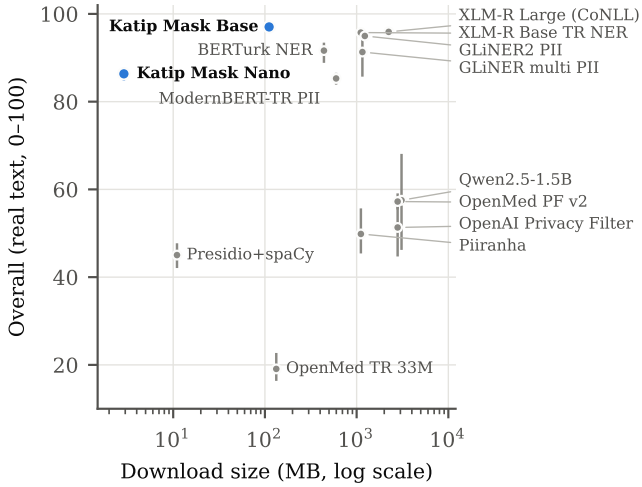


Figure 1: Overall (95% CI) against download size. Blue: Katip Mask models, both run in a browser.

5 Distribution: local masking in front of AI assistants

Katip Mask runs as a static web page (rules in Python via Pyodide, models via transformers.js) and as **Katip Kalkan**, a local MCP server packaged as a Claude plugin and a Claude Desktop extension (AGPL-3.0; the bundled Nano model is CC BY-SA 4.0). The assistant asks Kalkan for a file in a local folder; Kalkan reads and masks it on the device (rules plus Nano, with local OCR for scans and photos via PaddleOCR PP-OCRv5 in ONNX) and returns only text with placeholders such as `[AD SOYAD-1]`. The placeholder map stays on the device, and a restore tool writes the assistant's answer with the real values back to a local file, returning only its path. Two caveats matter. A file *attached* to the chat is uploaded with the message before any tool runs, so it must be opened through Kalkan instead. And on first start Kalkan downloads three Python packages (onnxruntime, numpy, cryptography) if `uv` is available; document processing itself makes no network calls. The connector's default detection threshold (0.7) is stricter than the benchmark setting (0.5), so its recall is not measured here.

System	Params	Size MB	Pages/s	**Overall** [95% CI]	TR names	Foreign	Prose	Leak % [95% CI]	Browser
Katip Mask Base (ours)	110M	111	1.7	**97.0** [96.5, 97.5]	**98.6**	96.3	88.9	6.9 [5.9, 8.1]	yes
XLM-R Large (CoNLL-en)	559M	2,235	2.3	95.9 [94.9, 96.7]	97.6	97.5	**89.9**	**3.4** [2.9, 3.9]	–
XLM-R Base Turkish NER	277M	1,110	5.7	95.7 [94.8, 96.6]	98.5	**98.2**	86.3	6.6 [5.6, 7.8]	–
GLiNER2 PII (person)	307M	1,228	1.2	95.0 [94.1, 95.8]	98.1	**98.2**	83.9	5.8 [4.8, 7.0]	–
BERTurk NER	110M	440	6.0	91.6 [88.8, 93.4]	97.4	91.7	77.1	8.1 [6.9, 9.4]	–
GLiNER multi PII	289M	1,156	1.1	91.3 [85.7, 95.5]	93.1	92.9	85.2	10.5 [5.6, 16.7]	–
Katip Mask Nano (ours)	2.5M	**2.9**	**50.0**	86.3 [84.8, 87.7]	93.3	93.1	57.8	18.5 [16.7, 20.4]	yes
ModernBERT-TR PII	149M	598	4.8	85.3 [83.9, 86.6]	98.5	96.5	39.1	21.6 [19.7, 23.6]	–
Qwen2.5-1.5B, prompted	1.5B	3,087	0.01	57.5 [46.2, 68.1]	56.7	55.1	47.6	41.1 [31.3, 52.1]	–
OpenMed Privacy Filter v2	1.4B	2,799	0.17	57.2 [55.4, 59.1]	57.0	54.7	48.3	36.4 [34.2, 38.2]	–
OpenAI Privacy Filter	1.4B	2,799	0.11	51.3 [44.7, 56.2]	57.0	57.2	27.6	52.5 [48.1, 58.3]	–
Piiranha v1	278M	1,113	2.9	49.8 [45.4, 55.7]	40.8	60.2	56.7	45.4 [40.0, 49.4]	–
Presidio + spaCy xx (baseline)	–	11	31.3	45.0 [42.1, 47.7]	38.9	43.6	60.4	31.7 [29.8, 33.4]	–
OpenMed PII Turkish 33M	33M	133	7.5	19.1 [16.4, 22.7]	11.0	17.3	33.4	73.0 [69.8, 75.6]	–

Table 1: KVKK-Bench v0.1.2, public split, real-text tracks. Overall and the three name columns are F2 (%); the name columns are per-track F2 (court-turkish, court-foreign, prose). Leak = share of gold characters left readable (lower is better). CIs: sentence-clustered bootstrap, 10,000 resamples. Size = what is downloaded to run (int8 ONNX plus rules for Katip; weights as loaded otherwise). Pages/s: Python on one Apple-silicon laptop; not an in-browser speed. Browser = run inside a web page via transformers.js/ONNX Runtime Web; a dash means not tested, not that it cannot run there. Numbers match the live leaderboard and the saved predictions.

6 Limitations and future work

- **No real client documents.** Court sentences with filled slots and Wikipedia are proxies. Contracts, petitions and e-mails from practice are the target of a consented pilot whose documents never leave the firm; private lawyer documents are not and will not be in the benchmark.
- **Names only on real text.** IDs, addresses and health data appear only in generated documents from the generator family our models train on, so every non-name figure above is optimistic and excluded from Overall.
- **In-domain advantage.** The prose track is the Wiki NER test split; Base and Nano were trained on its train split, and Nano on TurkishLegalNER's train split (the court tracks use its test split). Other models were trained on their own Turkish or multilingual NER data. A hidden test of 2025–26 Yargıtay/Danıştay decisions and AYM judgments with court-published names is under construction.
- **Slot filling** puts names where courts wrote "…", which a model could learn to exploit; the AYM real-name track is meant to check this.
- **One annotator**; inter-annotator agreement on 100 hidden documents is planned.
- **Speed** is measured in Python on one laptop. "Runs in the browser" means demonstrated, not timed.
- **OCR** is tested only on Katip's own scans and synthetic phone photos. Adding OCRTurk [36] as an OCR track awaits a licence from its authors.
- **Licences.** Both models are CC BY-SA 4.0 because they are trained on Wiki NER; code is AGPL-3.0. A v3 retrained without Wikipedia data (synthetic, court decisions, TurkishLegalNER) would allow Apache-2.0, if it stays within about one Overall point. Fine-tuning XLM-R on our data is queued.
- **Masking is not anonymisation.** Masked documents remain personal data under KVKK; the tool reduces exposure but does not by itself make an AI workflow compliant.

Reproducibility. `lab/foreign/build.py` (branch `main`) and `lab/legal/bench_build.py` (branch `legal-study`, commit `14fa7ab`) rebuild the public split from public sources, and `lab/legal/bench_run.py -rescore` re-scores the saved predictions; the bootstrap script and figure are in `paper/`. All numbers in this paper were reproduced from those files on 14 November 2026.

Ethics statement. No private or client documents were used. Every personal value inserted into the benchmark is synthetic; the only real names are those that courts or Wikipedia published. The benchmark measures leakage so that tools can be compared, not to certify compliance. Reporting follows Datasheets [9], Data Statements [2] and Model Cards [24] (Appendix B; model cards on Hugging Face).

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A Per-track results

B Datasheet summary

- **Motivation.** Compare Turkish personal-data maskers on the same text with the same scorer; built by one developer (Katip), unfunded.
- **Composition.** Public split v0.1.2: court-turkish 327 docs / 1,011 names; court-foreign 327 / 1,011; prose 1,000 / 654; generated tracks 380 docs (80 Katip documents in 16 styles, 300 foreign-client petitions) with TCKN, VKN, YKN, passport, foreign ID, IBAN, card, phone, e-mail, address, plate, birth date and health values; no-pii 3 docs. Language: Turkish (court register and Wikipedia); names from ten origin groups.
- **Collection and labelling.** Court text: TurkishLegalNER test split, original PER tags; slots refilled programmatically, so gold is exact by construction. Names: Wikidata (CC0) via QLever, filtered against every Katip lexicon and training set. Prose: Turkish Wiki NER test annotations as published. One annotator for checks; no crowd work.
- **Preprocessing.** Case suffixes re-inflected to the new name; casing sampled to match court style; titles kept outside spans; a canary string heads every file.
- **Uses.** Evaluating detectors before masking documents for AI tools. Not for training (the canary asks crawlers to exclude it) and not as evidence of legal compliance.
- **Distribution.** GitHub (scorer, builder, predictions) and the Hugging Face Space; licences per source (Apache-2.0, CC BY-SA 4.0, CC BY 4.0).
- **Maintenance.** Label errors and privacy requests via GitHub issues (privacy removals within 7 days); versioned changelog; every system is re-scored when labels change.
- **Known defects.** Listed in the limitations; the YKN/passport harness defect is reported in Section 4.

	court-turkish					court-foreign					prose				
System	R	P	Leak	Clean	FA/1k	R	P	Leak	Clean	FA/1k	R	P	Leak	Clean	FA/1k
Katip Mask Base	99.9	93.5	0.2	99	3.3	97.0	93.3	2.9	90	3.3	89.8	85.7	17.6	52	6.7
XLM-R Large (CoNLL-en)	99.1	92.2	0.1	97	4.1	99.1	91.7	0.2	95	4.3	89.3	92.6	9.8	59	3.2
XLM-R Base Turkish NER	98.3	99.4	1.1	94	0.3	97.9	99.4	1.2	90	0.3	85.6	89.2	17.7	49	4.7
GLiNER2 PII	98.7	95.7	1.8	94	2.1	98.7	96.4	1.8	93	1.8	91.1	63.6	13.8	57	23.0
BERTurk NER	96.9	99.5	2.2	87	0.2	89.9	99.6	4.7	69	0.2	74.5	89.6	17.3	47	4.0
GLiNER multi PII	93.3	92.4	7.3	96	3.6	92.8	93.4	7.7	94	3.1	88.2	75.1	16.6	56	12.5
Katip Mask Nano	95.0	87.1	5.6	80	6.9	94.8	87.0	5.4	79	6.8	57.5	58.9	44.5	28	19.2
ModernBERT-TR PII	98.9	97.1	1.9	76	1.5	96.2	97.7	3.2	65	1.3	34.3	91.3	59.6	11	2.2
Qwen2.5-1.5B, prompted	55.9	60.1	40.4	51	19.3	54.0	59.9	41.5	50	18.6	56.3	29.4	41.5	33	65.1
OpenMed Privacy Filter v2	56.1	60.9	35.2	32	25.6	53.4	60.3	35.2	31	27.3	52.9	35.9	38.8	28	54.3
OpenAI Privacy Filter	54.0	73.6	42.9	23	10.2	53.8	76.6	44.0	23	8.6	24.6	53.8	70.7	17	9.6
Piiranha v1	36.1	84.3	58.6	35	4.4	56.1	85.1	31.9	54	6.0	55.7	61.0	45.6	35	17.1
Presidio + spaCy xx	40.6	33.4	42.2	28	43.6	45.8	36.5	40.7	23	43.3	75.1	33.9	12.4	76	58.9
OpenMed PII Turkish 33M	9.8	21.2	85.9	1	46.3	15.8	27.9	76.7	7	45.7	35.0	28.1	56.5	14	49.4

Table 2: Recall (R), precision (P), leak rate, clean-document rate (documents with nothing left readable) and false alarms per 1,000 words (FA/1k), all in %, except FA/1k. Point estimates on the public split.